Reading: Active Calculus **Section 7.1**

Definition: If $F(x)$ and $f(x)$ are functions such that F is differentiable and $F'(x) = f(x)$, then we say that F is an **antiderivative** of f.

Activity. Use your knowledge of the rules of differentiation to propose antiderivativse for each of the following functions

(a) $f(x) = \sin(x)$

$f'(x) = -\cos x$

$\boxed{-\cos x + C}$

(b) $g(x) = \cos(x)$

$\boxed{\sin x + C}$

(c) $h(x) = e^x$

$\boxed{\int e^x = e^x}$

(d) $l(x) = m$ for some constant m

$\boxed{\int M = Mx + C}$

(e) $q(x) = x$

$\boxed{\int x = \frac{x^2}{2} + C}$

(f) $r(x) = x^2$

$\boxed{\int x^2 = \frac{x^3}{3} + C}$

(g) $s(x) = x^3$

$\boxed{\int x^3 = \frac{x^4}{4} + c}$

(h) $p(x) = x^n$ where n is an integer and $n \neq -1$

$\boxed{\int x^n = (n \neq -1)}$

(i) $z(x) = \frac{1}{x}$ where $x > 0$

$\boxed{\int \frac{1}{x} = \ln x + C}$